

## 7. Lipschitz maps

In this chapter we present some of the basic properties of Lipschitz maps and their relations to Hausdorff measures.

**7.1. Definition.** A map  $f: A \rightarrow \mathbf{R}^n$ ,  $A \subset \mathbf{R}^m$ , is a *Lipschitz map* if there is a constant  $L < \infty$  such that

$$|f(x) - f(y)| \leq L|x - y| \quad \text{for } x, y \in A.$$

The smallest such constant  $L$  is called the *Lipschitz constant* of  $f$  and denoted by  $\text{Lip}(f)$ .

### Extension of Lipschitz maps

Lipschitz maps can be extended:

**7.2. Theorem.** If  $f: A \rightarrow \mathbf{R}^n$ ,  $A \subset \mathbf{R}^m$ , is a Lipschitz map, there is a Lipschitz map  $g: \mathbf{R}^m \rightarrow \mathbf{R}^n$  such that  $f = g|_A$ .

*Proof.* We can extend every coordinate function  $f_i$  of  $f$  by the formula

$$g_i(x) = \inf \{f_i(y) + \text{Lip}(f_i)|x - y| : y \in A\}.$$

Then  $g = (g_1, \dots, g_n)$  is the required map. □

In the above proof  $\text{Lip}(g_i) = \text{Lip}(f_i)$ , whence  $\text{Lip}(g) \leq \sqrt{n} \text{Lip}(f)$ . It is much more difficult to show that one can actually find an extension  $g$  with  $\text{Lip}(g) = \text{Lip}(f)$ . This is called Kirszbraun's theorem, see Federer [3, 2.10.43].

### Differentiability of Lipschitz maps

We shall now prove Rademacher's theorem according to which Lipschitz maps are differentiable almost everywhere. If  $f: \mathbf{R}^m \rightarrow \mathbf{R}^n$  is differentiable at a point  $x \in \mathbf{R}^m$ , we denote by  $f'(x)$  its derivative at  $x$ ; it is a linear map  $\mathbf{R}^m \rightarrow \mathbf{R}^n$ . The following proof is from L. Simon [1].

**7.3. Theorem.** If  $f: \mathbf{R}^m \rightarrow \mathbf{R}^n$  is a Lipschitz map, then  $f$  is differentiable  $\mathcal{L}^m$  almost everywhere in  $\mathbf{R}^m$ .

*Proof.* By studying the coordinate functions we may assume  $n = 1$ . We shall consider the case  $m = 1$  to be known; the Lipschitz functions in one dimension are of bounded variation and their differentiability is studied in many text-books.

For  $e \in S^{m-1}$  and  $x \in \mathbf{R}^m$ , denote by  $\partial_e f(x)$  the derivative of  $f$  at  $x$  in the direction  $e$ , if it exists. Let  $B_e$  be the set of those  $x \in \mathbf{R}^m$  for which  $\partial_e f(x)$  does not exist. Then by simple standard arguments  $B_e$  is  $\mathcal{L}^m$  measurable. Applying the one-dimensional case to  $t \mapsto f(x + te)$ , we find for all  $x \in \mathbf{R}^m$  that

$$\mathcal{H}^1(B_e \cap \{x + te : t \in \mathbf{R}\}) = 0.$$

Hence by Fubini's theorem,  $\mathcal{L}^m(B_e) = 0$ . So we have shown that for any  $e \in S^{m-1}$ ,  $\partial_e f(x)$  exists for  $\mathcal{L}^m$  almost all  $x \in \mathbf{R}^m$ .

Next we prove that  $\partial_e f(x)$  is given by the gradient  $\nabla f(x) = (\partial_1 f(x), \dots, \partial_m f(x))$  as expected:

$$(1) \quad \partial_e f(x) = e \cdot \nabla f(x) \quad \text{for } \mathcal{L}^m \text{ almost all } x \in \mathbf{R}^m.$$

Here  $\partial_i f = \partial_{u_i} f$  are the partial derivatives in the standard basis  $\{u_i\}$ . Let  $\varphi \in C_0^\infty(\mathbf{R}^m)$ . By a change of variable we have for  $h \neq 0$ ,

$$\int h^{-1} [f(x + he) - f(x)] \varphi(x) dx = - \int h^{-1} [\varphi(x) - \varphi(x - he)] f(x) dx.$$

Since  $f$  is Lipschitz, we can let  $h \rightarrow 0$  and use Lebesgue's dominated convergence theorem and partial integration to obtain

$$\begin{aligned} \int \partial_e f(x) \varphi(x) dx &= - \int f(x) \partial_e \varphi(x) dx \\ &= - \int f(x) (e \cdot \nabla \varphi(x)) dx = - \sum_{j=1}^m e \cdot u_j \int f(x) \partial_j \varphi(x) dx \\ &= \sum_{j=1}^m e \cdot u_j \int \varphi(x) \partial_j f(x) dx = \int \varphi(x) (e \cdot \nabla f(x)) dx. \end{aligned}$$

Since this holds for all  $\varphi \in C_0^\infty(\mathbf{R}^m)$ , we get (1).

Let  $\{e_1, e_2, \dots\}$  be a dense subset of  $S^{m-1}$ . For each  $i$ , let  $A_i$  be the set of those  $x \in \mathbf{R}^m$  for which  $\nabla f(x)$  and  $\partial_{e_i} f(x)$  exist and  $\partial_{e_i} f(x) =$

$e_i \cdot \nabla f(x)$ . Denote  $A = \bigcap_{i=1}^{\infty} A_i$ . Then by what we have proven so far,  $\mathcal{L}^m(\mathbf{R}^m \setminus A) = 0$ .

We shall now show that  $f$  is differentiable at all points of  $A$ . For  $x \in A$ ,  $e \in S^{m-1}$  and  $h > 0$ , let

$$Q(x, e, h) = h^{-1}[f(x + he) - f(x)] - e \cdot \nabla f(x).$$

It suffices to show that for a given  $x \in A$ ,  $\lim_{h \downarrow 0} Q(x, e, h) = 0$  uniformly in  $e$ . First note that for  $e, e' \in S^{m-1}$ , with  $L = \text{Lip}(f)$ ,

$$|Q(x, e, h) - Q(x, e', h)| \leq (m+1)L|e - e'|.$$

Let  $\varepsilon > 0$ . By the compactness of  $S^{m-1}$  there is  $N \in \mathbf{N}$  such that if  $e \in S^{m-1}$  then  $|e - e_i| < \varepsilon/(2(m+1)L)$  for some  $i \in \{1, \dots, N\}$ . By the definition of  $A$ ,  $\lim_{h \rightarrow 0} Q(x, e_i, h) = 0$  for all  $i$ . Thus there is  $\delta > 0$  such that

$$|Q(x, e_i, h)| < \varepsilon/2 \quad \text{for } 0 < h < \delta, \quad i \in \{1, \dots, N\}.$$

Hence if  $e \in S^{m-1}$  and  $0 < h < \delta$ , we can choose  $i \in \{1, \dots, N\}$  with  $|e - e_i| < \varepsilon/(2(m+1)L)$  and obtain

$$\begin{aligned} |Q(x, e, h)| &\leq |Q(x, e, h) - Q(x, e_i, h)| + |Q(x, e_i, h)| \\ &< (m+1)L|e - e_i| + \varepsilon/2 < \varepsilon. \end{aligned}$$

This completes the proof.  $\square$

When  $m = 1$ , Rademacher's theorem is very precise: if  $A \subset \mathbf{R}$  with  $\mathcal{L}^1(A) = 0$ , there is a Lipschitz map  $\mathbf{R} \rightarrow \mathbf{R}$  which is not differentiable at any point of  $A$ , see Zahorski [1]. Rather surprisingly this is not true when  $m \geq 2$ . Preiss [5] has shown that there exists a Borel (even  $G_\delta$ ) subset  $A$  of  $\mathbf{R}^2$  such that  $\mathcal{L}^2(A) = 0$  and every Lipschitz function  $\mathbf{R}^2 \rightarrow \mathbf{R}$  is differentiable at some point of  $A$ .

The following strong approximation with  $C^1$  maps is often useful.

**7.4. Theorem.** *If  $f: \mathbf{R}^{\bar{m}} \rightarrow \mathbf{R}^n$  is a Lipschitz map and  $\varepsilon > 0$ , there is a continuously differentiable map  $g: \mathbf{R}^m \rightarrow \mathbf{R}^n$  such that*

$$\mathcal{L}^m(\{x : f(x) \neq g(x)\}) < \varepsilon.$$

We shall neither prove nor use this result. It can be proven with the help of Rademacher's theorem and Whitney's extension theorem for  $C^1$  maps, see Federer [3, 3.1.16].

The proof of the following theorem is a simple exercise based on the definition of  $\mathcal{H}^m$ .

**7.5. Theorem.** If  $f: \mathbf{R}^m \rightarrow \mathbf{R}^n$  is a Lipschitz map,  $0 \leq s \leq m$ , and  $A \subset \mathbf{R}^m$ , then

$$\mathcal{H}^s(fA) \leq \text{Lip}(f)^s \mathcal{H}^s(A).$$

In particular,

$$\dim(fA) \leq \dim A.$$

### A Sard-type theorem

We now prove a simple version of Sard's theorem for Lipschitz maps. A very general form can be found in Federer [3, 3.4.3]. For related results, see also Norton [1] and [2].

**7.6. Theorem.** If  $f: \mathbf{R}^m \rightarrow \mathbf{R}^n$  is a Lipschitz map, then

$$\mathcal{H}^m(\{f(x) : \dim(f'(x)\mathbf{R}^m) < m\}) = 0.$$

*Proof.* Let  $0 < R < \infty$  and set

$$A = \{x \in B(R) : \dim(f'(x)\mathbf{R}^m) < m\}.$$

Let  $\varepsilon > 0$  and  $L = \text{Lip}(f)$ . If  $x \in A$  and  $W_x = f'(x)\mathbf{R}^m + f(x) = \{f'(x)y + f(x) : y \in \mathbf{R}^m\}$ , then for sufficiently small  $r > 0$ ,

$$fB(x, r) \subset B(f(x), Lr) \cap \{y : d(y, W_x) \leq \varepsilon r\}.$$

Since  $\dim W_x \leq m - 1$  we have with some constant  $c$  depending only on  $m$  and  $n$ ,

$$\mathcal{H}_\infty^m(fB(x, r)) \leq c\varepsilon r(Lr)^{m-1}.$$

By Vitali's covering theorem 2.2 we can find disjoint such balls  $B_i = B(x_i, r_i)$  such that

$$\mathcal{L}^m\left(A \setminus \bigcup_{i=1}^{\infty} B_i\right) = 0 \quad \text{and} \quad \sum_{i=1}^{\infty} \mathcal{L}^m(B_i) < \mathcal{L}^m(A) + \varepsilon.$$

Then  $fA \subset (\bigcup_i fB_i) \cup f(A \setminus \bigcup_i B_i)$  and  $\mathcal{H}^m(f(A \setminus \bigcup_i B_i)) = 0$  by Theorem 7.5. Hence

$$\begin{aligned} \mathcal{H}_\infty^m(fA) &\leq \sum_{i=1}^{\infty} \mathcal{H}_\infty^m(fB_i) \leq cL^{m-1}\varepsilon \sum_{i=1}^{\infty} r_i^m \\ &\leq cL^{m-1}\alpha(m)^{-1}\varepsilon(\mathcal{L}^m(A) + \varepsilon), \end{aligned}$$

and the theorem follows letting  $\varepsilon \downarrow 0$  and recalling Lemma 4.6.  $\square$

### Hausdorff measures of level sets

We shall now study Hausdorff measures of level sets under Lipschitz maps.

**7.7. Theorem.** *Let  $A \subset \mathbf{R}^n$  and let  $f: A \rightarrow \mathbf{R}^m$  be a Lipschitz map. If  $m \leq s \leq n$ , then*

$$\int^* \mathcal{H}^{s-m}(A \cap f^{-1}\{y\}) d\mathcal{L}^m y \leq \alpha(m) \operatorname{Lip}(f)^m \mathcal{H}^s(A).$$

Here  $\int^*$  denotes the upper integral, recall Chapter 1.

*Proof.* We cover  $A$  for every  $k = 1, 2, \dots$  with closed sets  $E_{k,1}, E_{k,2}, \dots$  such that  $d(E_{k,i}) \leq 1/k$  and

$$\sum_i d(E_{k,i})^s \leq \mathcal{H}_{1/k}^s(A) + 1/k.$$

Let

$$F_{k,i} = \{y \in \mathbf{R}^m : E_{k,i} \cap f^{-1}\{y\} \neq \emptyset\}.$$

If  $y, z \in F_{k,i}$ , there are  $u, v \in A \cap E_{k,i}$  such that  $f(u) = y$  and  $f(v) = z$ . Then

$$|y - z| \leq \operatorname{Lip}(f)|u - v| \leq \operatorname{Lip}(f) d(E_{k,i}),$$

whence  $d(F_{k,i}) \leq \operatorname{Lip}(f) d(E_{k,i})$  and

$$\mathcal{L}^m(F_{k,i}) \leq \alpha(m) (\operatorname{Lip}(f) d(E_{k,i}))^m.$$

Using Fatou's lemma we obtain

$$\begin{aligned} & \int^* \mathcal{H}^{s-m}(A \cap f^{-1}\{y\}) d\mathcal{L}^m y \\ &= \int^* \lim_{k \rightarrow \infty} \mathcal{H}_{1/k}^{s-m}(A \cap f^{-1}\{y\}) d\mathcal{L}^m y \\ &\leq \int \liminf_{k \rightarrow \infty} \sum_i d(E_{k,i} \cap f^{-1}\{y\})^{s-m} d\mathcal{L}^m y \\ &\leq \liminf_{k \rightarrow \infty} \sum_i \int_{F_{k,i}} d(E_{k,i} \cap f^{-1}\{y\})^{s-m} d\mathcal{L}^m y \\ &\leq \liminf_{k \rightarrow \infty} \sum_i d(E_{k,i})^{s-m} \mathcal{L}^m(F_{k,i}) \\ &\leq \alpha(m) \operatorname{Lip}(f)^m \liminf_{k \rightarrow \infty} \sum_i d(E_{k,i})^s \\ &\leq \alpha(m) \operatorname{Lip}(f)^m \liminf_{k \rightarrow \infty} (\mathcal{H}_{1/k}^s(A) + 1/k) \\ &= \alpha(m) \operatorname{Lip}(f)^m \mathcal{H}^s(A). \end{aligned}$$

□

**7.8. Remarks.** The opposite inequality is in general false with any constants. For example, if  $A = E \times E$  where  $E \subset \mathbf{R}$  with  $\mathcal{H}^{s/2}(E) > 0$ ,  $s > 1$ , and  $\mathcal{L}^1(E) = 0$ , and  $f$  is the projection on the  $x$ -axis, then the left hand side is zero though  $\mathcal{H}^s(A) > 0$  (cf. Theorem 8.10). So there is no Fubini theorem for Hausdorff measures.

A more general inequality than that in Theorem 7.7 is proven in Federer [3, 2.10.25].

It is a relatively simple matter to show that the integrand in Theorem 7.7 is  $\mathcal{L}^m$  measurable if  $A$  is  $\mathcal{H}^s$  measurable and  $\mathcal{H}^s(A) < \infty$ . Indeed, for sets with  $\mathcal{H}^s(A) = 0$ , the integrand is zero almost everywhere by Theorem 7.7 and hence measurable. For compact sets a short argument is needed, and the rest follows by the approximation theorem 1.10. The integrand is also measurable for arbitrary Borel sets, but this seems to require the theory of Suslin sets, see Dellacherie [1] for a proof.

### The lower density of Lipschitz images

We shall now give a simple proof to show that a Lipschitz image of  $\mathbf{R}^m$  has positive  $m$ -dimensional lower density almost everywhere. Geometric properties of such Lipschitz images will be extensively studied from Chapter 15 on, and it will then be shown that the density even exists and equals one.

**7.9. Theorem.** *If  $f: \mathbf{R}^m \rightarrow \mathbf{R}^n$  is a Lipschitz map and  $A \subset \mathbf{R}^m$  is  $\mathcal{L}^m$  measurable, then  $\Theta_*^m(fA, x) > 0$  for  $\mathcal{H}^m$  almost all  $x \in fA$ .*

*Proof.* We may assume  $\mathcal{L}^m(A) < \infty$ . Let  $E = fA$ ,  $\varepsilon > 0$ , let  $C$  be a compact subset of  $E$  such that  $\Theta_*^m(E, x) < \varepsilon$  for  $x \in C$  and let  $U$  be open with  $A \subset U$  and  $\mathcal{L}^m(U) < \infty$ . We shall show that

$$\mathcal{H}^m(C) \leq c\varepsilon \mathcal{L}^m(U),$$

where  $c$  depends only on  $m$  and  $L$  with  $L = \text{Lip}(f)$ . We assume  $L > 0$ . Then the approximation theorem 1.10(1) implies

$$\begin{aligned} \mathcal{H}^m(\{x \in E : \Theta_*^m(E, x) = 0\}) &\leq \mathcal{H}^m(\{x \in E : \Theta_*^m(E, x) < \varepsilon\}) \\ &\leq c\varepsilon \mathcal{L}^m(A) \end{aligned}$$

for all  $\varepsilon > 0$ , and the theorem follows.

Because of the covering theorem 2.8 we can find disjoint closed balls  $B_i = B(x_i, r_i)$  and points  $y_i$ ,  $i = 1, 2, \dots$ , such that  $x_i \in C$ ,  $y_i \in A$ ,  $f(y_i) = x_i$ ,

$$\begin{aligned}\mathcal{H}^m(E \cap B_i) &< \varepsilon d(B_i)^m, \\ D_i = B(y_i, r_i/L) &\subset U \quad \text{and} \\ \mathcal{H}^m\left(C \setminus \bigcup_{i=1}^{\infty} B_i\right) &= 0.\end{aligned}$$

Then  $fD_i \subset B_i$ , whence the  $D_i$ 's are disjoint. Consequently,

$$\begin{aligned}\mathcal{H}^m(C) &= \sum_{i=1}^{\infty} \mathcal{H}^m(C \cap B_i) \leq \varepsilon \sum_{i=1}^{\infty} d(B_i)^m \\ &= c\varepsilon \sum_{i=1}^{\infty} \mathcal{L}^m(D_i) \leq c\varepsilon \mathcal{L}^m(U),\end{aligned}$$

as required.  $\square$

The proof of Theorem 7.9 is from Martin and Mattila [2] where an analogous result is proved for Hölder continuous maps.

### Remarks on Lipschitz maps

**7.10.** We comment here on some other interesting measure-theoretic properties of Lipschitz maps. For many others see Federer [3].

(1) It is easy to see that any  $\mathcal{L}^1$  measurable subset  $A$  of  $\mathbf{R}$  with  $\mathcal{L}^1(A) > 0$  can be mapped with a Lipschitz map onto an interval. To do this we may assume  $A \subset [0, \infty)$  and define  $f$  by

$$f(x) = \mathcal{L}^1(A \cap [0, x]) \quad \text{for } x \in A.$$

The corresponding question in higher dimensions is much more difficult. Recently Preiss [6] proved that if  $A$  is an  $\mathcal{L}^2$  measurable subset of  $\mathbf{R}^2$  with  $\mathcal{L}^2(A) > 0$ , there is a Lipschitz map  $f: A \rightarrow \mathbf{R}^2$  such that  $fA$  is a disc. In  $\mathbf{R}^n$  for  $n \geq 3$  the problem is unsolved.

(2) Let  $A$  be a compact subset of  $\mathbf{R}^2$  such that  $\mathcal{H}^1(A) > 0$ . Is there a Lipschitz map  $f: A \rightarrow \mathbf{R}$  such that  $\mathcal{L}^1(fA) > 0$ ? In general the answer is negative; a counterexample is constructed by Vitushkin, Ivanov and Melnikov [1]. It even satisfies the regularity condition

$$r/c \leq \mathcal{H}^1(A \cap B(x, r)) \leq cr \quad \text{for } x \in A, \ 0 < r \leq 1.$$

For some sets like self-similar sets satisfying the open set condition, recall 4.13, such a map is fairly easy to construct.

**7.11. Bi-Lipschitz maps.** A map  $f: A \rightarrow B$ ,  $A \subset \mathbf{R}^m$ ,  $B \subset \mathbf{R}^n$ , is a *bi-Lipschitz map* if  $f$  is Lipschitz and it has Lipschitz inverse  $f^{-1}: B \rightarrow A$ . Here are some recent measure-theoretic results on them.

(1) The following was proved by Jones [1] generalizing an earlier result of David [3]. For any positive integers  $m$  and  $n$  and  $\varepsilon > 0$  there exists an integer  $N(\varepsilon)$  such that if  $Q$  is the unit cube of  $\mathbf{R}^n$ ,  $m \geq n$ , and if  $f: Q \rightarrow \mathbf{R}^m$  with  $\text{Lip}(f) = 1$ , then there are  $B_1, \dots, B_N \subset Q$ ,  $N \leq N(\varepsilon)$ , such that

$$\mathcal{H}_\infty^n\left(f\left(Q \setminus \bigcup_{i=1}^N B_i\right)\right) < \varepsilon$$

and each  $f|_{B_i}$  is bi-Lipschitz with  $\text{Lip}((f|_{B_i})^{-1}) \leq N(\varepsilon)$ .

(2) Buczolic [1] proved that a bi-Lipschitz map  $f: A \rightarrow B$ , where  $A, B \subset \mathbf{R}^n$  are  $\mathcal{L}^n$  measurable, preserves every Lebesgue density point; for almost every such point this is easy.

(3) Falconer and Marsh [2] and [3] studied the question of which self-similar or certain related sets are bi-Lipschitz equivalent. Obviously such sets must have the same Hausdorff dimension, and inside some classes this also suffices.

(4) Saaranen (unpublished) proved that if  $0 < s < t < 1$ ,  $E \subset \mathbf{R}^n$  is a compact set such that for some  $0 < a < b < \infty$ ,

$$ar^s \leq \mathcal{H}^s(E \cap B(x, r)) \leq br^s \quad \text{for } x \in E, \quad 0 < r \leq 1,$$

then  $E$  is bi-Lipschitz equivalent with some subset of  $C(\lambda)$  where  $C(\lambda)$  is the Cantor set of 4.10 with  $\dim C(\lambda) = t$ . Evidently, for topological reasons, this fails for  $s \geq 1$ .

### Exercises.

1. Let  $F$  be a closed subset of  $\mathbf{R}^n$ . Show that  $x \mapsto d(x, F)$  is a Lipschitz function.
2. Construct a Lipschitz function  $f_\lambda: [0, 1] \rightarrow \mathbf{R}$  which is not differentiable at the points of  $C(\lambda)$ , where  $C(\lambda)$  is as in 4.10. (See also Darst [1] for the differentiability properties of the usual Cantor function.)

A map  $f: A \rightarrow \mathbf{R}^n$ ,  $A \subset \mathbf{R}^m$ , is *Hölder continuous* with exponent  $\alpha$ ,  $0 < \alpha \leq 1$ , if there is  $L < \infty$  such that

$$|f(x) - f(y)| \leq L|x - y|^\alpha \quad \text{for } x, y \in A.$$



3. Modify Theorem 7.2 and its proof to Hölder continuous maps.
4. Generalize Theorem 7.5 to Hölder continuous maps.
5. Construct a Hölder continuous parametrization  $f: [0, 1] \rightarrow K$  for the von Koch curve  $K$ , recall 4.13, which is not differentiable at any point of  $[0, 1]$ . What can be said about the Hölder exponent of  $f$ ?
6. Show that the Minkowski and packing dimensions cannot increase under Lipschitz maps. Derive also estimates for Hölder continuous maps.